



1. Data Modeling (or Data Model Design).

Data Modeling is the process of organizing data into related tables and defining the relationships between them to support efficient data analysis, reporting, and decision-making.

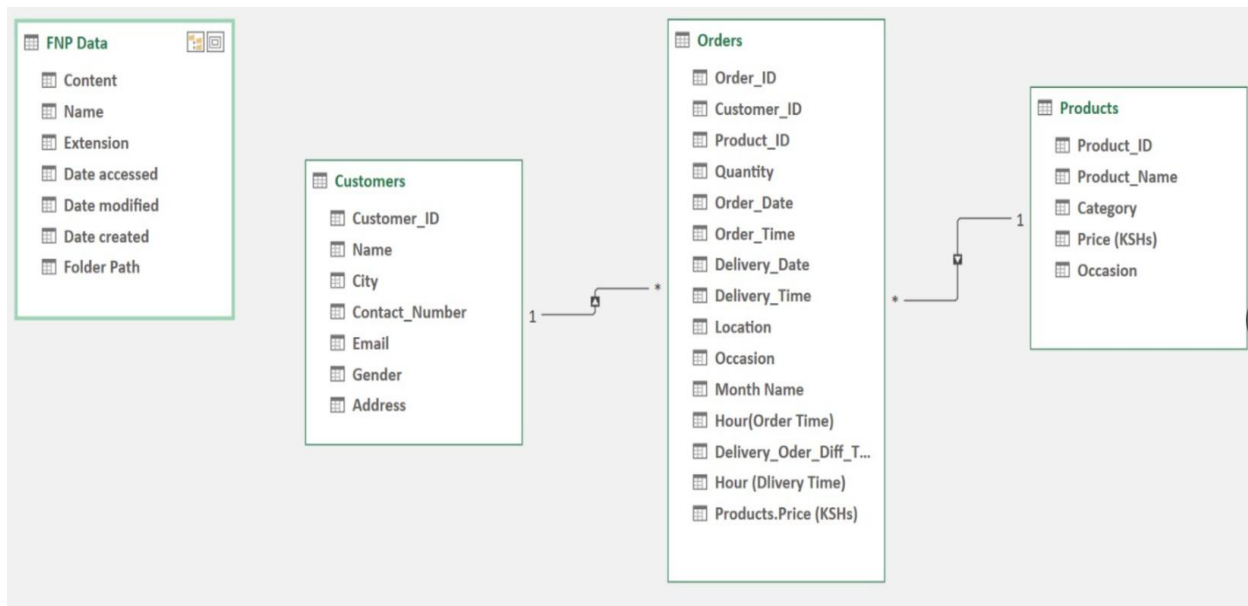
In the Context of this Project

The data modeling process involved:

- Identifying the required data entities (Customers, Orders, and Products).
- Creating separate tables for each entity.
- Defining relationships between the tables using common fields such as **Customer_ID** and **Product_ID**.
- Establishing one-to-many relationships to enable accurate analysis and reporting.

Data modeling was carried out to structure the dataset into related tables and establish relationships among them. The Customers, Orders, and Products tables were linked through unique identifiers, creating a relational model that supports efficient data analysis, reduces redundancy, and improves dashboard performance.

2. Data Model and Relationships





A data model is a structured framework that organizes data into tables and defines the relationships between them. In dashboard projects, a data model helps ensure that information from different tables can be connected, analyzed, and visualized accurately. It serves as the foundation upon which reports, charts, and Key Performance Indicators (KPIs) are built.

Description of the Data Model Used

FNP Sales Trends Analysis project uses a **relational data model** consisting of three main tables: **Customers**, **Orders**, and **Products**. These tables are connected through unique identifiers known as **primary keys** and **foreign keys**.

- **Customers Table** stores customer information such as Customer ID, Name, City, Contact Number, Email, Gender, and Address.
- **Orders Table** contains transaction details including Order ID, Customer ID, Product ID, Quantity, Order Date, Delivery Date, Location, Occasion, and delivery-related metrics.
- **Products Table** stores product information such as Product ID, Product Name, Category, Price, and Occasion.

Relationships Between Tables

The tables are linked using **one-to-many (1:M) relationships**:

1. Customers → Orders

- Connected through the **Customer_ID** field.
- One customer can place multiple orders.
- Each order belongs to only one customer.

2. Products → Orders

- Connected through the **Product_ID** field.
- One product can appear in multiple orders.
- Each order references a specific product.

These relationships allow the dashboard to combine customer, order, and product information seamlessly for analysis.



Importance of the Data Model

The data model plays a critical role in the dashboard because it:

- Eliminates data duplication and redundancy.
- Ensures consistency and accuracy of reports.
- Enables efficient filtering and cross-analysis across tables.
- Supports calculations such as total sales, revenue, order counts, and delivery performance.
- Improves dashboard performance and data management.

Data Model Structure

Customers (1) ----- (*) Orders (*) ----- (1) Products

Where:

- represents one record.
- (*) represents many records.

The data model forms the backbone of the dashboard by linking customer, order, and product data through defined relationships. This structure enables effective analysis of sales performance, customer behavior, product trends, and operational efficiency, thereby supporting informed decision-making through interactive dashboard visualizations.